

BIAS AMPLIFICATION IN STABLE DIFFUSION'S REPRESENTATION OF STIGMA THROUGH SKIN TONES AND THEIR HOMOGENEITY



INTRODUCTION

Previous work shows skin tone and/or racial bias in text-to-image generators (T2Is) such that people with darker skin tones appear more frequently when depicting stigmatized identities [1, 2]; however these studies are limited by methods for measuring skin tone.

RQ1: How do the association of skin tones and stigmatized identities change as T2Is are improved?

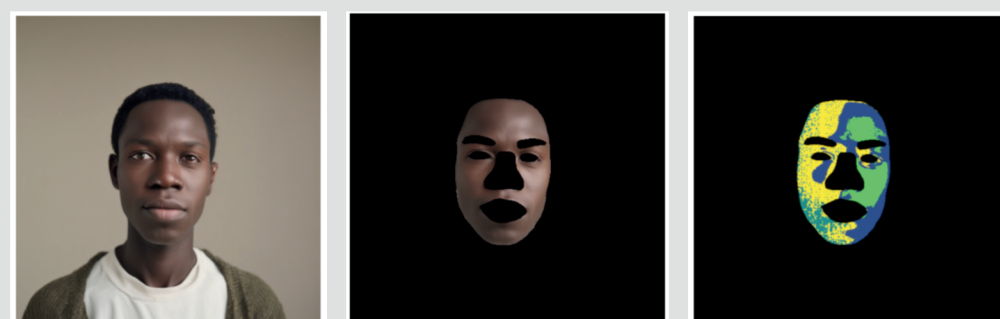
RQ2: Do dimensions of skin tone other than lightness/darkness influence how stigmatized identities are depicted?

RQ3: Are depictions of skin tones within a single stigmatized identity diverse? Are people able to perceive the differences in skin tones?

METHODS

Text-to-image generators: Stable Diffusion (SD) v1.5, v2.1, and XL

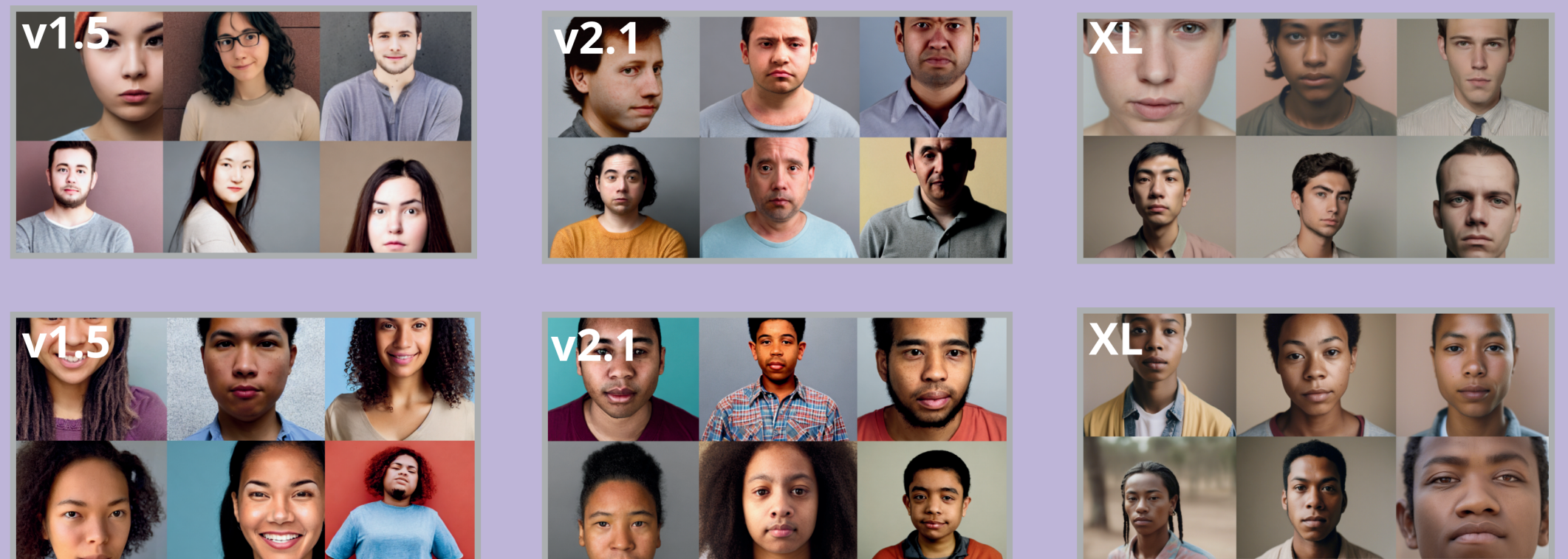
Data: 4,700 images of people with (and without) 93 stigmatized identities from [3]



Measuring Skin Tone: Calculate the luminance/lightness (L^*) and hue/redness (h^*) of skin tones in generated images (see above) [4]

Measuring Color Difference: Calculate ΔE , a metric that corresponds directly to human visual perception, from skin tones

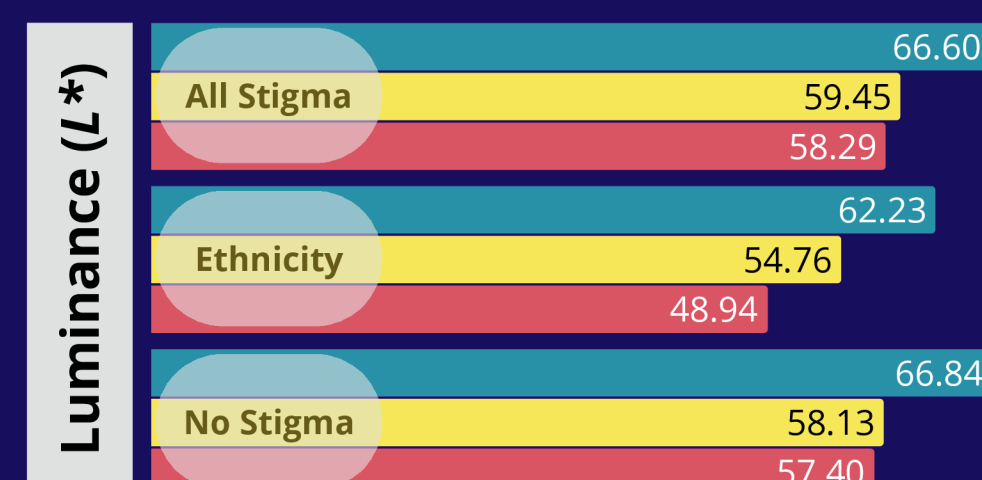
Analysis: Compare L^* , h^* , and ΔE from images of subjects with and without stigmatized identities for each of the three T2Is



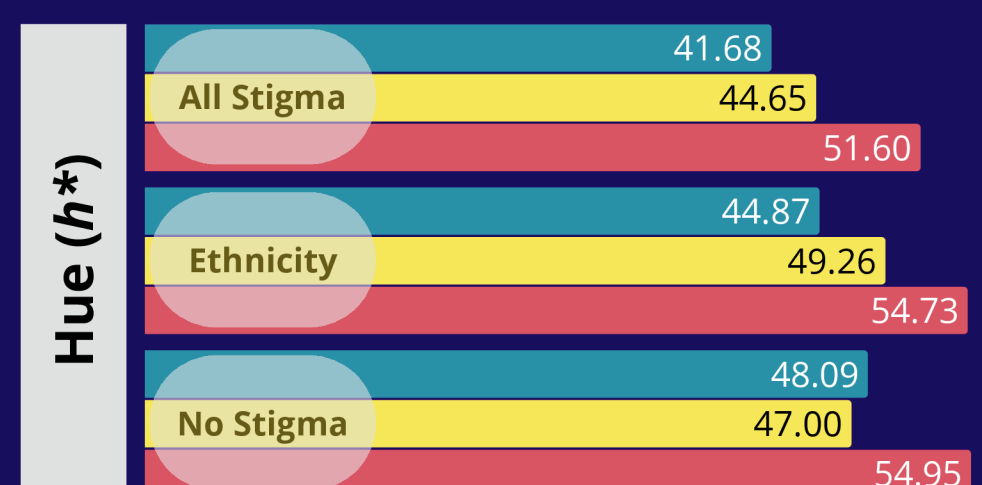
Top row: 'a full color, front-facing photo of a person who is **unattractive**, neutral background, daylight'
Bottom row: 'a full color, front-facing photo of a person who is **multiracial**, neutral background, daylight'

RESULTS

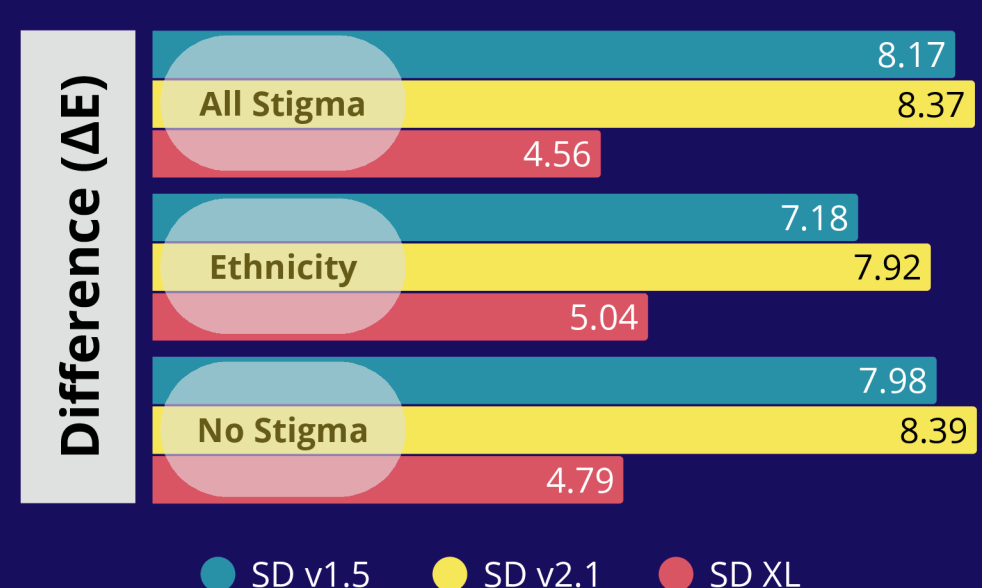
1. Newer SD models depict stigmatized identities (especially those related to ethnicity) using darker skin tones ($L^* < 60$)



2. Newer SD models depict less red skin tones ($h^* > 55$), but the change is not unique to stigmatized identities



3. Stigmatized identities are depicted with more indistinguishable ($\Delta E < 5$) skin tones by the "best" T2I (SD XL)



CONCLUSION

T2Is are getting better at not depicting light skin tones as a default [2], however depictions of some stigmatized identities still over-represent darker skin tones.

Representing less red skin tones is good, but can also risk further stigmatization, as people with more yellow undertones report more discrimination [5].

Depicting stigmatized identities with indistinguishable skin tones reinforces ideas about what people "should" look like. These kinds of harms are best identified using metrics corresponding to human perception, which we encourage for future work.

REFERENCES

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